What Mathematical Work Do AI Tutors Leave for Learners? Prompting and Decision Routing in TutorMoments

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Abstract: AI tutors must support learners without completing the reasoning they need to practice. We examine this tension using TutorMoments-Preview. A qualitative analysis of 38 matched moments codes the mathematical work left by plain and evaluation-aware responses. In the first AI-assisted coding pass, evaluation-aware responses invited connections, explanations, or strategies in 16/38 cases, versus 2/38 plain responses. A separate experiment generates 1,560 first replies with Qwen3-32B across all 520 benchmark moments. Archived Jev and OpenAI Decisions outputs select between matched scaffold and rigor replies. Under automated evaluation, pass rates rose from 68.7% for detailed self-choice to 72.9% and 73.5%, but fixed rigor guidance reached 74.0%. Jev's gain shrank to 0.9 percentage points in the strict matched mathematics frame. The findings distinguish category agreement from response execution and show why routing claims require fixed-policy controls and inspection of the work left for learners.

Problem and theoretical grounding

A mathematically correct tutor response can still do the work a learner needs to practice. It may supply the next calculation, explain the decisive relationship, or choose a strategy before the student has a chance to contribute. The assistance dilemma concerns how much support to provide and when to withhold it (Koedinger & Aleven, 2007). Research on teacher questioning similarly directs attention to how questions elicit and develop students' mathematical thinking (Franke et al., 2009). For AI tutoring, this raises a design question: what remains for the learner to do after the tutor responds?

TutorMoments-Preview operationalizes this tension through teacher-identified situations calling for scaffolding or a push for rigor (Zhang et al., 2026). Explicit instructions about the trade-off improve the benchmark's tutoring scores, but neither a situation label nor the presence of a question tells us exactly what mathematical work remains. A tutor can ask a question after supplying its answer. Conversely, a hint can make a demanding task accessible while preserving the learner's reasoning. Scaffolding and rigor can also occur together: the benchmark permits a generated exchange classified as both to satisfy either reference category, provided it does not over-scaffold.

We connect two analyses of this design problem. First, a qualitative comparison examines the work left by archived plain and evaluation-aware responses. Second, a technical extension asks whether a separate decision model can guide a small tutor more effectively than detailed instructions alone. StratL steers a tutor through expert-defined teaching intents (Puech et al., 2025), and DICE separates intervention choices from response generation (Pal et al., 2026). Eduardo trains smaller tutors with reinforcement learning and evaluates them on TutorMoments (Macina et al., 2026). We test an inference-time alternative: a frozen binary router with matched self-selected and fixed guidance, examining whether any improvement requires a router. We ask: (1) How do tutoring instructions change the mathematical work left for learners? (2) Does explicit decision routing improve a small model's response relative to self-selected and fixed guidance, and what overhead does it introduce?

Data and qualitative analysis

TutorMoments-Preview contains 462 de-identified transcripts of remote human tutoring with students in grades 2–7, including spoken dialogue and descriptions of shared-screen activity. Experienced teachers identify instructional moments and describe the situation, tutor action, and result; a language model maps these annotations to task categories, which are combined by majority vote. The published balanced evaluation uses 520 moments, with 260 scaffolding and 260 rigor conditions (Zhang et al., 2026). We used the released histories and continuations without collecting new classroom data. For qualitative analysis, we locked 40 situations from an initial 370-situation mathematics comparison frame. Excluding students used in earlier pilots left 226 eligible situations. Seeded sampling selected 20 per category and one per student. A subsequent speech/event audit reduced the overall matched frame to 346, requiring usable first replies, clear turn boundaries, and complete source-response availability. It also excluded two locked cases because additional learner events preceded the observed human reply. Neither was replaced. The final sample comprised 38 students and 38 moments, evenly divided between scaffolding and rigor situations. Each contributed the human continuation and released GPT-5.5 continuations under

plain and evaluation-aware prompts. The primary comparison therefore involved 76 AI response blocks forming 38 matched pairs; 38 human blocks provided a naturalistic reference, not a third controlled prompt condition.

A response block combines consecutive spoken tutor utterances and ends at the next learner utterance or observable action. Screen descriptions and pauses were retained in the context but excluded from spoken response text. Following a theory-informed content-analysis approach (Hsieh & Shannon, 2005), we developed two separate codes. *Work Left* (W) identifies the most substantive unsupplied mathematical demand: no new work or repetition (W0), a specified calculation or heavily cued recognition (W1), a connection or explanation within a narrowed path (W2), or a strategy, argument, or diagnosis with meaningful discretion (W3). *Information Supplied* (R) describes what the tutor has already provided, from no answer information (R0) to the complete active answer (R3). Unclear cases receive NA. These are study-specific descriptions, not validated scales of teaching quality; more remaining work is not always appropriate for a particular learner.

We report the first archived AI-assisted coding pass, conducted on source-masked responses with preceding context. Each judgment includes a rationale and literal evidence excerpt. A second archived pass is retained as a sensitivity analysis. All 114 response texts and evidence quotations in the reported pass were checked against their source packets. The full codebook, both code sets, sampling record, and exclusions accompany the paper.

Technical extension: choosing a direction before replying

The technical analysis uses all 520 released benchmark moments, with 260 reference labels of each type. It is separate from the 38-case qualitative sample. We also report a sensitivity analysis on the strict matched mathematics frame described above; this subset differs in response eligibility as well as subject matter. The full conversation before each stopping point is retained. Teacher labels, annotation hints, future continuations, and learner personas are excluded from tutor and router inputs.

Prior decision-only tests asked Jev 1.14.0 (Almeida, 2026) and OpenAI Decisions with GPT-6 Luna (OpenAI, n.d.) to choose scaffolding or rigor. Their strongest archived configurations matched 376/520 (72.3%) and 386/520 (74.2%) reference labels, respectively. A later prompt revision reduced these scores to 367 and 372. We freeze the earlier configurations for the extension and preserve all runs in the supporting materials. These are exploratory, reused-benchmark results: previous errors informed prompt development, so they are not held-out generalization estimates. Three pairs share the same context and student metadata but have opposite reference labels, illustrating ambiguity in treating the category as a unique next-action target.

All response arms retain the publisher's evaluation-aware prompt, with additional next-reply guidance. For the response experiment, Qwen3-32B (Qwen Team, 2025) generates three first replies at every moment: one that selects its own direction under detailed instructions, one given scaffold guidance, and one given rigor guidance. The scaffold and rigor branches form a paired set of potential responses. Cached Jev and Decisions choices select the corresponding branch afterward, allowing both routers, fixed-scaffold and fixed-rigor policies, and a reference-label policy to be compared on identical generated responses. This isolates branch selection from generation variation. The reference-label policy is a diagnostic control and is never an input to either router. It is not an upper bound on pedagogical quality.

Gemini 2.5 Flash receives the context and randomly ordered replies without condition names or reference labels. Primary response success requires the assigned-direction behavior, no detected over-helping or mathematical error, full context fit, and valid generation. Replies expressing both directions can satisfy either condition. A secondary rate uses direction and over-helping alone. A fixed, balanced subset of 32 distinct students is also assessed by Claude Sonnet 4.6. We retain disagreement between evaluators and use paired comparisons with student-cluster resampling. The prompts, sample, providers, and analysis are frozen before the final judgments.

This is a first-response evaluation, not the publisher's multi-turn replay with a simulated learner. Its scores cannot be ranked directly against the website leaderboard. Generator time to first answer is measured from streamed calls; cached routing costs and timings are reported separately. Adding historical router duration gives an estimated serial first-answer time, not an observed end-to-end pipeline latency. All Qwen generation calls use a fixed provider, temperature zero, non-thinking mode, and a 384-token output cap; incomplete or invalid replies count as failures.

Findings

Explicit instructions changed the work left

Evaluation-aware responses more often left an unsupplied mathematical action (Table 1). Sixteen of 38 invited a connection, explanation, or strategy-level contribution (W2/W3), compared with two plain responses. Complete

active answers (R3) appeared in 21/38 plain responses and 5/38 evaluation-aware responses; one plain response was unassessable for R. The codes capture different features: supplying an answer and inviting an explanation can coexist, so R3 does not by definition imply W0.

Table 1: Work left in the first archived coding pass ($n = 38$ per column).

Work left	Plain	Evaluation-aware	Human reference
W0: No new mathematical work or repetition	21	2	20
W1: Specified calculation or cued recognition	15	20	12
W2: Connection or explanation	2	14	1
W3: Strategy, argument, or diagnosis	0	2	0
NA: Unclear active demand	0	0	5

Case E01 illustrates a modest but consequential difference. For $25 \times 14 = 25 \times (2 \times \star)$, the plain response states, "So the star should be 7," and verifies the equality (W0/R3). The evaluation-aware response asks, "What number times 2 makes 14?" (W1/R2). It supplies the decisive relationship but leaves the calculation. This is not automatically deep reasoning, but it preserves a contribution that the plain reply completes.

The second pass preserved the direction of the comparison but assigned W2/W3 to five plain and 23 evaluation-aware responses. In E22-B, a tutor states "84 seven times" and asks for multiplication notation. One pass treated this as cued recognition (W1); the other treated it as connecting a story to a representation (W2). The comparison is consistent across passes, but these boundary judgments affect its magnitude.

Routing and response quality

Detailed self-choice passed 357/520 cases (68.7%), Jev-directed selection 379 (72.9%), and Decisions-directed selection 382 (73.5%; Table 2). Jev's paired gain was 4.2 percentage points (95% student-cluster bootstrap interval: -1.0 to 9.2); Decisions' gain was 4.8 points (0.2 to 9.5). Neither exceeded fixed rigor guidance (385/520, 74.0%). Jev improved rigor-condition passes from 149 to 180 but reduced scaffolding-condition passes from 208 to 199. Thus, its aggregate improvement concealed a trade-off across situations.

Table 2: Passing first replies under the primary evaluator.

Policy	All 520	Scaffold 260	Rigor 260	Matched 346
Detailed self-choice	68.7%	80.0%	57.3%	72.0%
Jev-directed	72.9%	76.5%	69.2%	72.8%
Decisions-directed	73.5%	80.8%	66.2%	75.1%
Always scaffold	56.5%	86.2%	26.9%	59.8%
Always rigor	74.0%	65.8%	82.3%	76.0%
Reference-label diagnostic	84.2%	86.2%	82.3%	86.4%
Opposite-label diagnostic	46.3%	65.8%	26.9%	49.4%

The last column uses the strict matched mathematics frame. The two label-based policies are diagnostic controls, not deployable systems. All policies share the same paired scaffold/rigor replies, except self-choice.

The stricter mathematics comparison was less favorable to routing: Jev passed 252/346 cases versus 249 for self-choice, a 0.9-point difference (-5.3 to 6.5). Decisions passed 260 and fixed rigor 263. On the 32-case evaluation audit, Claude passed 16 self-choice replies and 22 each for Jev, Decisions, and fixed rigor. On those same cases, the primary evaluator passed 20 self-choice and 19 Jev replies, reversing the relative result. Overall agreement was 75/96 replies (78.1%): condition differences were evaluator-sensitive.

Category agreement and reply success diverged. When Jev matched the reference category, 320/376 selected replies passed; when it disagreed, 59/144 still passed. Across the paired policy replies, both passed at 214 moments, only rigor passed at 171, only scaffolding at 80, and neither at 55. Even reference-label selection passed only 438/520. Better category prediction therefore leaves a separate response-execution problem, while category disagreement does not invariably produce a poor reply. In moment 24, for example, the learner was working on $1{,}000 - 10$. Both routers correctly selected scaffolding, but the selected reply returned to an earlier $338 + 586$ problem; the evaluator marked a context failure.

Median generator time to first answer was 1.20 seconds for self-choice. Adding archived routing time yielded reconstructed medians of 1.43 seconds for Jev and 1.60 for Decisions. Estimated selected-response costs

per 1,000 cases were $0.59, $0.90, and $1.25, respectively. These estimates exclude evaluation and do not establish a deployment speedup. The complete new experiment cost approximately $3.00, including evaluation and conservative failed-call reservations.

Discussion and implications

The qualitative and technical analyses address different parts of the same instructional decision. W/R coding asks what the learner can still contribute after a reply; a router chooses a broad direction before a reply exists. Agreement with a situation label therefore cannot substitute for examining the generated response. A scaffold can preserve a difficult inference, while a request for explanation can merely ask the learner to repeat reasoning already supplied.

Our technical results provide limited support for adding a binary router. Both learned selectors improved the full-preview point estimate relative to self-choice, but fixed rigor guidance matched or exceeded them, and Jev's gain nearly disappeared in the strict matched mathematics frame. This does not imply that tutors should always demand rigor. The fixed-rigor replies also expressed scaffolding in many cases, and the evaluator recognized both behaviors in 202/520 replies under that policy. A policy name is therefore an imperfect description of the support actually delivered. The useful design target may be selecting a specific next demand and the information to withhold, rather than treating scaffolding and rigor as mutually exclusive response types. The paired branch analysis makes this distinction inspectable without attributing differences to separate sampled replies.

For designers, the relevant comparison is whether a separate decision step improves responses beyond equally detailed self-selection and simple fixed guidance. A gain over a weak prompt would not isolate the value of routing. Likewise, low router latency alone does not establish a faster system, because a sequential router adds a request before generation. The practical goal is to preserve accessible mathematical work while controlling total response cost and delay.

The study is limited by purposive benchmark construction, exploratory use of previously inspected data, AI-assisted coding, and automated response evaluation. The human continuations are naturalistic references; the generated branches are not observed interventions with learners. Neither analysis measures student uptake, retention, or learning gains. The next validation step is independent human coding and evaluation on previously unseen students and moments, followed by studies of what learners actually do with the resulting support. The present contribution is a reproducible account of opportunities in tutor text and a controlled test of whether explicit routing improves first-response tutoring behavior.

Supporting materials and disclosure

The course submission folder contains the codebook, source-linked coding data, both coding passes, exclusions, frozen experiment instructions, response-level results, and reproducibility files. AI tools assisted coding, implementation, response evaluation, and drafting; no independent human coding validation is claimed. Original dataset labels and all completed experimental conditions are retained.

References

Almeida, D. (2026, September 15). *Introducing system one models & Jev*. TypeSafe AI. https://typesafe.ai/blog/introducing-system-one-models-and-jev

Franke, M. L., Webb, N. M., Chan, A. G., Ing, M., Freund, D., & Battey, D. (2009). Teacher questioning to elicit students' mathematical thinking in elementary school classrooms. *Journal of Teacher Education*, *60*(4), 380–392. https://doi.org/10.1177/0022487109339906

Hsieh, H.-F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative Health Research*, *15*(9), 1277–1288. https://doi.org/10.1177/1049732305276687

Koedinger, K. R., & Aleven, V. (2007). Exploring the assistance dilemma in experiments with cognitive tutors. *Educational Psychology Review*, *19*(3), 239–264. https://doi.org/10.1007/s10648-007-9049-0

Macina, J., Kapur, M., & Sachan, M. (2026). *The assistance dilemma: Learning to teach via multi-turn reinforcement learning* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2610.06446

OpenAI. (n.d.). *Decisions* [API documentation]. Retrieved November 11, 2026, from https://developers.openai.com/api/docs/guides/decisions

Pal, S., Ji, K., & Srihari, R. K. (2026). *DICE: Decoupling capability from intervention necessity in LLM tutoring* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2610.04825

Puech, R., Macina, J., Chatain, J., Sachan, M., & Kapur, M. (2025). Towards the pedagogical steering of large language models for tutoring: A case study with modeling productive failure. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Findings of the association for computational linguistics: ACL 2025* (pp. 26291–26311). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.findings-acl.1348

Qwen Team. (2025). *Qwen3-32B* [Model card]. Hugging Face. Retrieved November 11, 2026, from https://huggingface.co/Qwen/Qwen3-32B

Zhang, A., Ross, A., Patel, K., Bernado, J., Bowie, R., Ribeiro, A. T., Halper, D., Valayaputtur, H., Andreas, J., Loeb, S., Li, L., Lo, K., & Knight, R. (2026). *When help is unhelpful: Evaluating AI tutors for productive struggle* [Working paper]. Allen Institute for AI. https://tutormoments.allen.ai/static/paper/tutormoments-preview.pdf